Q4

The network has 3 fully connected layers with the architecture:

$$784 \rightarrow 512 \rightarrow 512 \rightarrow 10$$

The input image is $28 \times 28 = 784$ pixels, and the output layer has 10 neurons.

Total number of parameters:

Layer	Weight parameters	Bias parameters	Total
`fc1`: $784 \rightarrow 512$	$784 \times 512 = 401,408$	512	401,920
`fc2`: $512 \rightarrow 512$	$512 \times 512 = 262,144$	512	262,656
`fc3`: $512 \rightarrow 10$	$512 \times 10 = 5,120$	10	5,130
Total	668,672	1,034	669,706

Answer: 669,706 total parameters.

Total operations per input image:

According to the question:

- 1 MAC = 2 operations
- 1 ReLU activation = 1 operation
- 1 bias addition = 1 operation

Layer	MACs	MAC operations	Bias additions	ReLU operations	Total operations
`fc1`	401,408	802,816	512	512	803,840
`fc2`	262,144	524,288	512	512	525,312
`fc3`	5,120	10,240	10	10	10,260
Total	668,672	1,337,344	1,034	1,034	1,339,412

Answer: 1,339,412 operations per input image.